

# Ellen Poulsen

 Ellen Poulsen |  [ecpoulse@calpoly.edu](mailto:ecpoulse@calpoly.edu) |  +1 (541) 654-2129

## EDUCATION

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2025–Present      **B.S. Aerospace Engineering** (GPA: 3.66/4.0)  
California Polytechnic State University, San Luis Obispo

2023      Recipient of the Oregon State Seal of Bilingualism in Spanish

**Relevant Coursework:** Introduction to Design and Manufacturing, Welding, Statics & Mechanics, Stars & Planetary Systems, Linear Analysis (expected Spring 2027), Dynamics (expected Spring 2027)

## WORK EXPERIENCE

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**Test Engineering Intern — Empirical Systems Aerospace** *June 2026 — Aug. 2026*  
*San Luis Obispo, CA*

- Developed comprehensive servo testing procedures, physical test stand, and automated test software.
- Programmed servo control mechanisms utilizing both PWM and CAN bus communication protocols.
- Calibrated data acquisition timing across multiple testing devices to ensure precise sensor telemetry collection.
- Managed project version control and code collaboration using Python, VS Code, and GitLab.

**Observatory Assistant — Cal Poly Physics Department** *May 2026 — Present*

- Operating 14-inch Meade LX600 ACF Telescope for public viewing events.
- Led educational observing sessions for student groups, presenting key astrophysics concepts and managing night-sky targets.

## PROJECTS

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**Cal Poly Space Systems — Liquids Project**

- Development of bipropellant liquid rocket engines, test stand, and associated test procedures and safety
- Familiarization with Nitrous Oxide and Ethanol as propellants, high pressure systems, and high temperature systems
- Static pressure testing, hot fires, cold flows, and work on various projects including regenerative rocket engine and test stand maintenance

## LEADERSHIP EXPERIENCE

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### **Cal Poly Space Systems Lead — SILLE (Student Introductory Liquids Learning Engine)**

- Guiding students through project to design and manufacture their own small liquid rocket engine to test up to 50lbs thrust
- Organization of over 40 students into teams, teaching basics of liquids rocketry and foundational understanding of isentropic equations
- Public speaking and overseeing safety operations for project
- Mentorship with focus on member retention and transfer of knowledge
- Proficient in communication utilizing Microsoft teams, powerpoint, forms, and excel

### **Society of Women Engineers at Cal Poly**

- Advocating for and supporting women underrepresented in fields of engineering.
- Community outreach and education opportunities to advance access to STEM and inspire the next generation of girls in engineering